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Project Title	Order Management System
Week	Week 4

WEEK-4 TASK

Task Name

Order Management System

Description

The **Order Management System** is designed to manage customer orders and the products included in each order. It uses **Orders** and **Order_Details** tables to store order information, quantities, dates, and total amounts. The system supports inserting and modifying orders and generating customer order history reports.

Objectives

1. Design **Orders** and **Order_Details** tables.
2. Manage customer product orders efficiently.
3. Store order date, quantity, and total amount.
4. Perform order insertion and modification operations.
5. Generate customer order history reports.

Tables

1. Orders

- order_id – Primary Key
- customer_id – Identifies the customer
- order_date – Date of the order

- total_amount – Total value of the order

2. Order_Details

- order_detail_id – Primary Key
- order_id – Foreign Key
- product_id – Identifies the product
- quantity – Number of products ordered
- unit_price – Price of one product
- subtotal – Quantity × Unit Price

SQL Implementation

```
CREATE DATABASE OrderManagementDB;
```

```
USE OrderManagementDB;
```

```
CREATE TABLE Orders (  
    order_id INT PRIMARY KEY,  
    customer_id INT,  
    order_date DATE,  
    total_amount DECIMAL(10,2)  
);
```

```
CREATE TABLE Order_Details (  
    order_detail_id INT PRIMARY KEY,  
    order_id INT,  
    product_id INT,  
    quantity INT,  
    unit_price DECIMAL(10,2),  
    subtotal DECIMAL(10,2),
```

```
FOREIGN KEY (order_id) REFERENCES Orders(order_id)
);
```

```
INSERT INTO Orders VALUES
(101,1,'2026-08-17',1500.00),
(102,2,'2026-08-17',2500.00);
```

```
INSERT INTO Order_Details VALUES
(1,101,201,2,500.00,1000.00),
(2,101,202,1,500.00,500.00);
```

```
UPDATE Orders
SET total_amount = 1800.00
WHERE order_id = 101;
```

```
UPDATE Order_Details
SET quantity = 3, subtotal = 1500.00
WHERE order_detail_id = 1;
```

```
SELECT o.order_id, o.customer_id, o.order_date,
       od.product_id, od.quantity, od.subtotal
FROM Orders o
JOIN Order_Details od
ON o.order_id = od.order_id
WHERE o.customer_id = 1;
```

Expected Output

Order ID	Customer ID	Order Date	Product ID	Quantity	Unit Price	Subtotal
101	1	2026-08-17	201	2	500.00	1000.00
102	1	2026-08-17	202	1	500.00	500.00
103	1	2026-08-17	204	2	500.00	1200.00

Conclusion

The **Order Management System** provides an organized way to store and manage customer orders. The relationship between Orders and Order_Details helps maintain order information, product quantities, prices, and customer order history efficiently.